

Global Climate Change Report 2026

Annual Assessment by the International Climate Research Council

Executive Summary

Global average temperatures have risen by 1.2°C above pre-industrial levels as of 2026. The past decade has been the warmest on record, with 2024 and 2025 breaking consecutive heat records. Arctic sea ice extent has declined by 13% per decade since satellite records began in 1979. Sea levels are rising at 3.7mm per year, a rate 2.5x faster than the 20th century average.

1. Key Climate Indicators

1.1 Temperature Trends

The global mean surface temperature anomaly for 2025 was +1.45°C relative to the 1850-1900 baseline. Land surfaces are warming at twice the rate of ocean surfaces. The Arctic is warming 4x faster than the global average, a phenomenon known as Arctic amplification.

- 2025 was the hottest year ever recorded globally
- 15 of the 16 hottest years occurred after 2010
- Ocean heat content reached record highs in all major basins
- Permafrost thaw releasing significant methane stores

1.2 Extreme Weather Events

The frequency and intensity of extreme weather events has increased significantly. Economic losses from climate-related disasters reached \$380 billion in 2025. Category 4 and 5 hurricanes have increased by 25% over the past 30 years. Droughts affecting food production now impact 40% more agricultural land than in 1990.

2. Emissions Data

Region	CO ₂ Emissions (GtCO ₂)	% of Global Total	Change Since 2000
China	11.47	31.8%	+189%
USA	4.71	13.1%	-21%
EU-27	2.53	7.0%	-34%
India	2.88	8.0%	+248%
Rest of World	14.41	40.1%	+67%

3. Projected Scenarios

Under current policies, global warming is projected to reach 2.7°C by 2100. To limit warming to 1.5°C, global emissions must be halved by 2030 and reach net-zero by 2050. Every fraction of a degree matters — at 2°C, 37% of the world's population faces severe heat stress, compared to 14% at 1.5°C.

4. Recommended Actions

- Transition to 100% renewable energy by 2040
- Phase out coal power plants by 2030 in developed nations
- Implement carbon pricing mechanisms globally
- Protect and restore 30% of land and ocean ecosystems
- Increase climate adaptation funding to \$300 billion/year